



OPEN FatigueNet: A hybrid graph neural network and transformer framework for real-time multimodal fatigue detection

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Fatigue creates complex challenges that present themselves through cognitive problems alongside physical impacts and emotional consequences. FatigueNet represents a modern multimodal framework that deals with two main weaknesses in present-day fatigue classification models by addressing signal diversity and complex signal interdependence in biosignals. The FatigueNet system uses a combination of Graph Neural Network (GNN) and Transformer architecture to extract dynamic features from Electrocardiogram (ECG) Electrodermal Activity (EDA) and Electromyography (EMG) and Eye-Blink signals. The proposed method presents an improved model compared to those that depend either on manual feature construction or individual signal sources since it joins temporal, spatial, and contextual relationships by using adaptive feature adjustment mechanisms and meta-learned gate distribution. The performance of FatigueNet outpaces existing benchmarks according to laboratory tests using the MePhy dataset to detect fatigue levels across four different categories during training and testing. The end-to-end latency performance of FatigueNet on MePhy benchmark reaches 50 ms per 20 s window on conventional hardware which proves its capability for real-time fatigue monitoring and surpasses baseline models.

Keywords Biological signals processing, Fatigue recognition, Graph neural networks (GNNs), Multimodality, Transformers

Fatigue is an essential biological signal that warns the body it should rest and recover when needed¹. It is a multifaceted phenomenon that involves physical and mental fatigue². Physical fatigue comes about from being worn out by strenuous physical work, but mental fatigue is the product of too much mental work – homework, or even deciding something³. Fatigue is a complicated condition that is further complicated by emotional factors like chronic sadness or depression. The perspective of fatigue from a neuroscience point of view is that there are physiological and neural alterations⁴. For example, fatigue is neuropathologically linked to reductions in the prefrontal cortex (responsible for decision-making and focus) and dysregulation in dopamine and serotonin systems (which govern mood and motivation)⁵. In addition, fatigue has physiological manifestations, including decreased muscle efficiency, abnormal cardiovascular state, and abnormal respiratory patterns^{6,7}. Fatigue is a pervasive condition, with adverse effects on people's daily lives, both cognitively and physiologically, and emotionally⁸. Fatigue decreases concentration, retention of memory, and learning capacity for students⁹. Fatigue can escalate in high-stakes environments, such as driving, to life-threatening scenarios, reducing reflexes and increasing the likelihood of accidents¹⁰. Wearable health monitoring devices have revolutionized how fatigue is assessed and managed recently. Physiological signals, such as heart rate variability (HRV) and peripheral blood flow, are measured by these devices, and provide important information about autonomic nervous system activity¹¹. A promising technique for detecting stress, a key biomarker for fatigue stratification, is photoplethysmography (PPG), a noninvasive, low-cost technique to analyze these signals. In high-risk environments, such as clinical environments or operational tasks, PPG allows for the detection of fatigue with high precision by measuring changes in vascular tone, cardiac rhythms, and sympathetic parasympathetic balance¹². Fatigue detection is further facilitated by the rapid growth of machine learning (ML) and deep learning (DL) technologies¹³. For example, the analysis of biosignals including electroencephalogram (EEG), electromyogram (EMG), and PPG has been performed with traditional ML methods, such as Support Vector Machines (SVM), Decision Trees, and Random Forests¹⁴. However they often depend on manual feature

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extraction, which restricts their ability to handle the complex and non-linear patterns in physiological data¹⁵. Figure 1 shows how the rapid growth of telemedicine has spurred the demand for real-time health monitoring wearable devices in healthcare management¹⁶. Detecting fatigue requires telehealth devices to analyze health-related signals recorded from bodily functions. Telehealth devices provide important physiological information that validates health management strategies for mental and physical wellness¹⁷.

The authors initially sought to show how standard Convolutional Neural Networks (CNNs)/Long short-term memory (LSTM) methods¹⁸, (good at spatial/temporal feature learning) compare with their hybrid, Graph Neural Networks (GNN) + Transformer approach despite their weakness in capturing distant inter-modal dependencies in fatigue signals^{19,20}. In addition, attention-based models, e.g. Transformers have shown their ability to focus on important intervals in physiological signals, improving interpretability and classification accuracy^{21,22}. However, these advancements rely on large labeled datasets, which are costly and time-consuming to acquire²³. Practical alternatives to supervised learning have emerged in the form of unsupervised and semi-supervised learning approaches for fatigue classification with little reliance on labeled data²⁴. In addition, GNNs have recently been able to model complex relationships between variables and time points, and their innovations make them ideal for fatigue classification tasks²⁵. Inspired by these developments, we propose FatigueNet, a novel multimodal model that leverages GNNs and Transformers to make progress in fatigue classification. Dynamic representations are used to extract meaningful features from the heterogeneity of multimodal biosignals, including ECG, Electrodermal Activity (EDA), EMG, and eye blinking data, in FatigueNet. FatigueNet integrates MetaNet as its last module to establish modality-specific relevance through meta-learning while the current context determines signal stream weighting before performing the final classification stage. The weights from the Meta-Gated Adaptive Fusion (MGAF) block integrate into the stage-wise aggregation and attention-based refinement step for low-latency context-sensitive feature fusion before the final classification process. The combination of MetaNet with MGAF makes FatigueNet able to automatically adapt its operations during runtime when signal quality or task requirements change. The key contributions of this research are as follows:

1. The proposal introduces a dual architecture which utilizes both a GNN along with a Transformer encoder for effective fusion of inter-modal features to boost fatigue classification from various biosignals.
2. FatigueNet delivers real-time low-latency fatigue detection on the MePhy benchmark dataset through which it performs better than traditional benchmarks by more than 5% in streaming conditions.
3. Our meta-learning based fusion module (MGAF) performs dynamic stream weights calculation to guarantee real-time functional capabilities along with an operational range spanning different operational settings.

This manuscript is structured as follows: Section II covers the datasets and methods used. The experimental results are presented in Section III, and the interpretation of the results, limitations, and directions for future work are discussed in Section IV. In Section V, the findings are summarized and implications are discussed.

Method & materials

In this section, we explain the dataset preparation process in detail and the FatigueNet pipeline architecture

Dataset

In this study, we use a multimodal benchmark dataset called MePhy to investigate fatigue induced by biological signals collected under randomized controlled trials at the University of Modena and Reggio Emilia, Italy²⁶. The dataset encompasses four primary biosignals: Electrocardiogram (ECG), EDA, EMG, and Eye Blinking Frequency. These signals were collected from 60 participants, 30 male and 30 female, with a mean age of 22.85

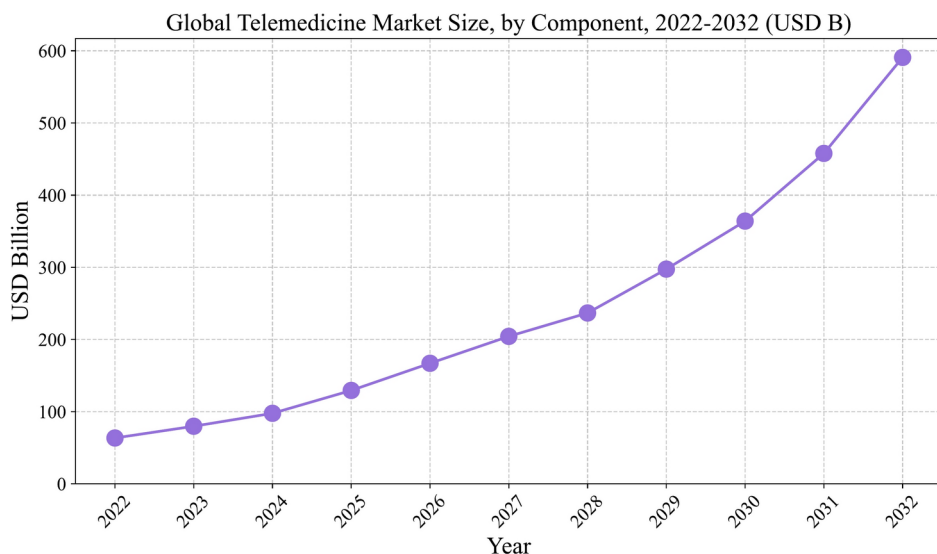


Fig. 1. Market revenue and growth forecast for the global telemedicine market.

years. The participants were subjected to four experimental conditions: rest and a state of mental and physical fatigue, mental fatigue, physical fatigue. The experimental protocol was ecologically valid by the use of tasks to induce fatigue in ecologically valid situations. Stroop tests, math calculations, and 2 back memory tasks were combined to induce mental fatigue, and physical fatigue was induced by isometric exercise with varying elbow flexion angles. The final condition combined fatigue types to simulate real-world multitasking challenges. The dataset included cognitive fatigue, physical fatigue, combined fatigue, and resting state (no fatigue) for each participant. Table 1 presents an overview of the dataset’s comprehensive nature, and the availability of biosignals across participants.

Pre-processing

We first downsample the ECG signal from the Polar H10’s 1 Hz sampling rate to match other measurements in ECG signal preprocessing. We address baseline wander in the EDA and EMG signals obtained at 1000 Hz through the BITalino kit by applying a wavelet transform to decompose the signal into its frequency components and remove low-frequency noise²⁷. A bandpass filter with a low pass cutoff of 500 Hz is used in the EMG processing to remove high-frequency noise while retaining the muscle signal. For motion artifacts and low-frequency disturbances, a high pass filter cutoff of 20 Hz is used to remove these while preserving the fidelity of the muscle signal²⁸. We filter out high-frequency noise using a low pass filter with a cutoff frequency of around 30 Hz for EDA. EDA and EMG signals are filtered with a notch filter at 50 Hz to remove powerline noise²⁹. Furthermore, to remove motion artifacts from the EMG signal, a median filter method was applied, and to remove outliers, a Z-score-based threshold was used to identify and replace outliers that could have been caused by significant recording errors or noise³⁰. Also, Interpolation methods are applied to fill in missing data. Optical flow analysis of video frames tracks eyelid movements captured at 30 Hz by the Logitech C920 camera³¹. This method computes blink frequency with pixel movement detection, without any additional filtering. Baseline drift and high-frequency noise were removed with a bandpass filter (0.5–30 Hz) and residual motion and muscle artifacts were removed with wavelet decomposition. Z scores were used to detect outliers and spikes which were replaced with interpolated values to remove artifacts. Signals were normalized to a consistent range for uniform analysis. Peak detection was used to identify blink events, with invalid durations (i.e., outside 100–400 ms) excluded, and each blink was segmented and centered within epochs to capture its full dynamics³². As well as, higher frequency signals were downsampled to the lowest sampling rate of all collected data to standardize signal processing across modalities. This step preserved critical information without compromising uniform temporal alignment³³. Figure 2 shows the preprocessing step for an example participant.

Methodology

The proposed approach integrates a FatigueNet architecture to tackle real-time fatigue recognition using multimodal biometric signals. This section outlines the model’s key components, workflows, and modeling details. Figure 3 illustrates the overall structure of the proposed model.

FatigueNet architecture

FatigueNet is a framework for predicting fatigue levels from multimodal analysis of biometric data (ECG, EMG, EDA, and eye blink). The raw data is used after conventional preprocessing methods such as filtering, outlier removal, peak detection, normalization, and downsampling. In the next step, feature engineering on the data is performed using the capabilities of GNN and transformers combined with MGAF so that our classifier, which is a Multiclass Support Vector Machine (MSVM), can accurately identify fatigue into four classes: low, medium, high and ultra.

Feature extraction

Several studies have demonstrated that for tasks such as stress and fatigue recognition, handcrafted features, like those described, can achieve performance levels comparable to deep features in scenarios where training data is limited. While deep learning excels with large-scale datasets, handcrafted features offer an interpretable and efficient alternative, particularly for applications involving transfer learning or individual-specific model adaptation. We used a sliding window method with a time window of 20 s and a step size of 5 s to extract features suitable for classification. The time and frequency domain features were accurately represented using this approach. FatigueNet utilizes a robust multimodal feature extraction framework to effectively analyze biosignals, including ECG, EMG, EDA, and eye-blinking data, across various domains: temporal, frequency, time-frequency, chaotic, and fractal domains. In the time domain, features extracted include ECG Heart Rate (HR), Heart Rate Variability (HRV), R-wave Amplitude (RA), Normal-to-Normal (NN) intervals, the Standard Deviation of NN Intervals (SDNNI), the Root Mean Square of Successive Differences (RMSSD), the Standard Deviation of the Difference between Adjacent NN Intervals (SDANNI), and the Percentage of mean NN Intervals greater than 50 (pNNI50) and 20 ms (pNN20). Additionally, EMG features such as Mean Absolute

Participant group	ECG	EDA	EMG	Eye blinking
Users 0–18	✓	✓	✓	✓
Users 19–29	✓	✓	✓	×
Users 30–59	✓	×	×	×

Table 1. The mephy dataset for each participant²⁶.

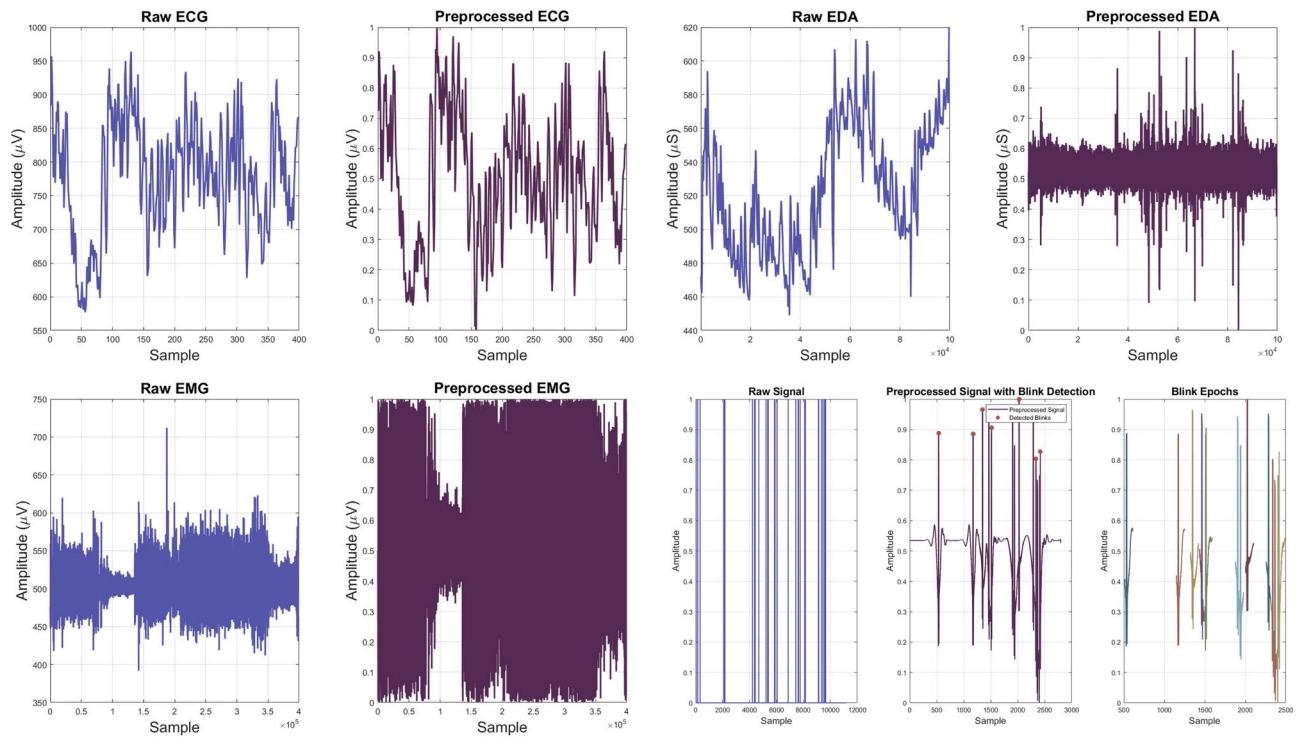


Fig. 2. Display cognitive fatigue data before and after preprocessing for participant 1.

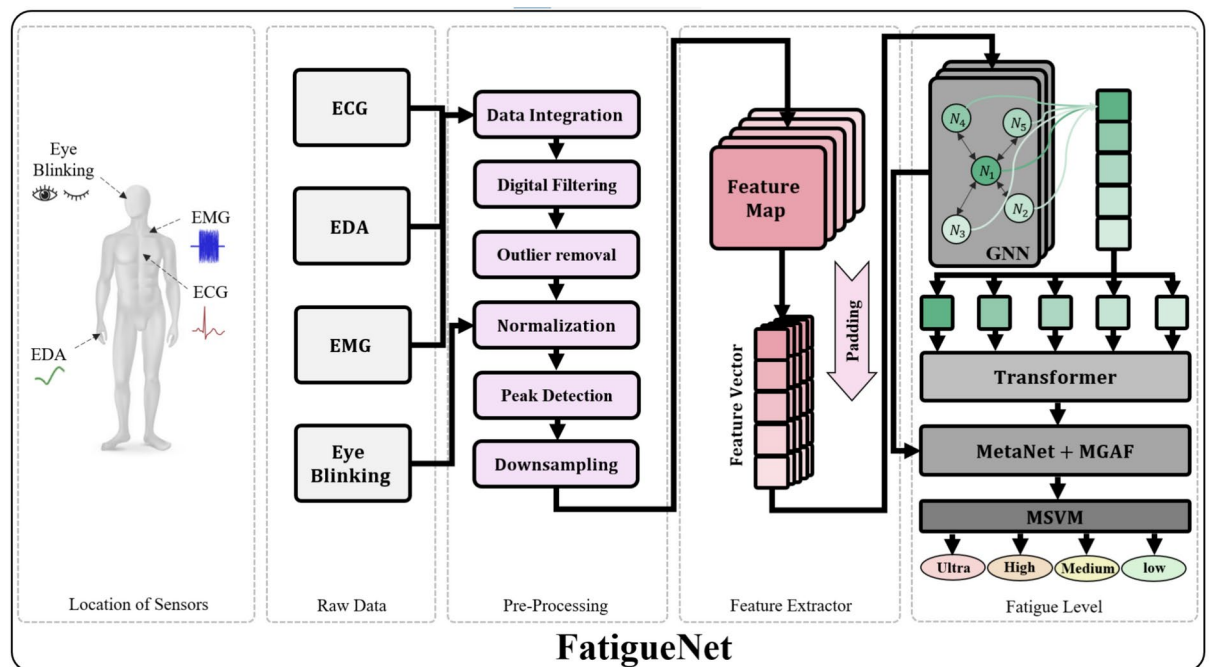


Fig. 3. The overall block diagram of the FatigueNet.

Value (MAV), Zero Crossing (ZC) count, Slope Sign Changes (SSC), and the Root Mean Square (RMS) of EMG are computed. EDA metrics like Galvanic Skin Responses (GSR) and Skin Conductance Level (SCL), along with blink duration measures for eye signals, are also included. In the frequency domain, features encompass ECG Power Spectral Density (PSD) across frequency bands, Ultralow Frequency (ULF), Very Low Frequency (VLF), Low Frequency (LF), High Frequency (HF), and the energy ratio of Low-to-High Frequency (LF/HF). Additional frequency-domain features include EMG Mean Frequency (MNF), Median Frequency (MDF), EDA

spectral patterns indicative of stress, and spectral analysis of blinking to detect fatigue. Chaotic features, such as Lyapunov exponents approximate entropy for ECG, and sample entropy for EMG, offer insights into signal complexity and irregularity. Continuous Wavelet Transform (CWT) to uncover non-stationary patterns and transient signal changes in ECG and EMG^{12,34,35}. Furthermore, fractal features include metrics such as the fractal dimension and Hurst exponent, which quantify the self-similarity and complexity of ECG and EMG signals. The use of this comprehensive approach allows for highly accurate temporal and spatial modeling of fatigue. Finally, feature sets from all modalities are integrated into a joint vector:

$$F_{\text{total}} = \{F_{\text{ECG}}, F_{\text{EMG}}, F_{\text{EDA}}, F_{\text{Blink}}\} \quad (1)$$

where $F_{\text{total}} \in \mathbb{R}^n$, and n is the dimensionality after concatenation. The feature vector is mapped into a high-dimensional feature space using CNN. The convolutional layers serve to extract spatial and hierarchical patterns from multimodal data:

$$\text{Feature Map}_{i,j} = \text{ReLU} \left(\sum_k W_k * F_{\text{total}} + b \right) \quad (2)$$

where ReLU (Rectified Linear Unit) is a non-linear activation function employed in CNNs, W_k is the convolution kernel, $*$ denotes the convolution operation, and b is the bias term. The structure of the feature extraction block of the proposed model is shown in Fig. 4.

GNN

The GNN module is designed to capture interactions between multimodal signals. Nodes represent individual feature maps, and edges denote intermodal dependencies:

$$h_v^{(k+1)} = \sigma \left(\sum_{u \in N(v)} \frac{1}{c_{v,u}} W \cdot h_u^{(k)} + b \right) \quad (3)$$

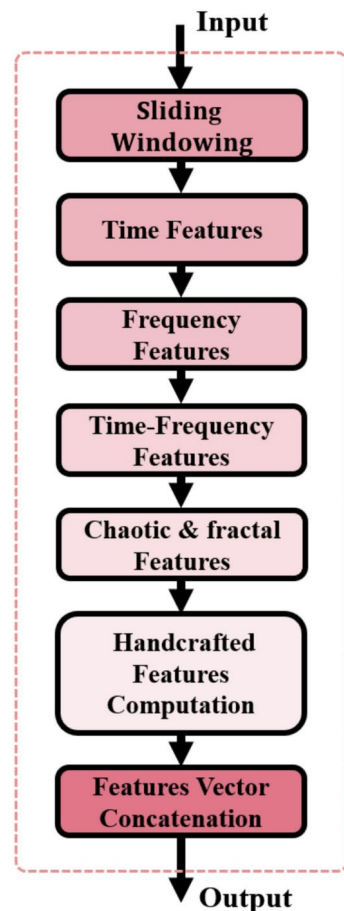


Fig. 4. Structure of the feature extraction block.

where $h_v^{(k)}$ represents the node embedding at layer k , $N(v)$ denotes neighbors of node v , and $c_{v,u}$ normalizes contributions. and W and b are learnable parameters. The activation function σ ensures the non-linear propagation of node features. As Well as, based on Fig. 3, in GNN, nodes N correspond to distinct physiological sensors (e.g., ECG, EDA). The adjacency matrix A is dynamically defined based on correlation coefficients between signal pairs:

$$A_{ij} = \frac{\text{corr}(X_i, X_j)}{\sum_{k \neq i} |\text{corr}(X_i, X_k)|} \quad (4)$$

This normalization strengthens the correlations and reduces the effect of redundant or weaker connections, allowing graph topology adjustment that is adaptive and data-driven. In addition, the custom GNN of this study employs graph filters of different sizes to extract temporal and spatial features. The filtering process is expressed as follows:

$$G_{c,f}^{(\ell)} = \sum_{k=1}^K g_{c,f,k}^{(\ell)} A^k \quad (5)$$

Specifically, A^k denotes the k hop neighborhood connectivity, and $g_{c,f,k}$ are learnable coefficients. The model also incorporates multi-scale filtering to preserve hierarchical dependencies between short- and long-range connections, thereby enriching the representation of inter-modal relationships. To address variability in feature distributions across datasets, each node's feature vector is normalized using L2 normalization:

$$x_c^{(\ell)} = \frac{x_c^{(\ell)}}{\|x_c^{(\ell)}\|_2 + \epsilon} \quad (6)$$

Where ϵ prevents numerical instability. This normalization process makes the learning robust and consistent, making this feature imbalance problem less important. The outputs of graph convolutions are activated using the ReLU function to introduce nonlinearity:

$$x_f^{(\ell+1)} = \text{ReLU} \left(\sum_{c=1}^{C^\ell} G_{c,f}^{(\ell)} x_c^{(\ell)} + b_f \cdot I \right), \quad (7)$$

b_f is the learnable bias. This transformation amplifies non-linear dependencies, so that the model represents complex interactions between modalities. Finally, attention-enhanced convolution is used to aggregate multi-hop neighbor contributions for custom GNN. The process utilizes query (Q), key (K), and value (V) matrices derived from node features:

$$x_f^{(\ell+1)} = \sigma \left(\text{softmax}(QK^T) V \right) \quad (8)$$

The proposed approach focuses on relevant neighbor interactions and thus improves the model's capacity to learn hierarchical and temporal dependencies. The overall block diagram of the GNN of the proposed model is shown in Fig. 5.

Transformer

A sequence of feature maps is processed by a Transformer module to capture long- term temporal dependencies in fatigue patterns. This mechanism adaptively adjusts the edge weights by combining learned parameters with positional priors (Fig. 6):

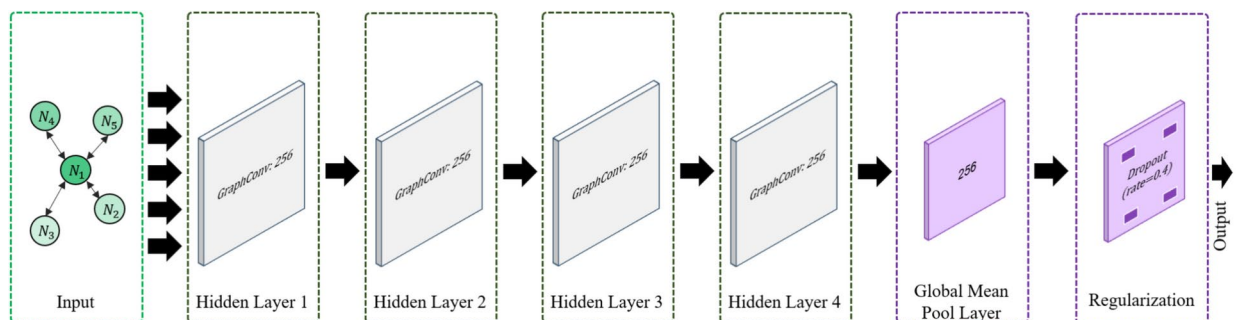


Fig. 5. The GNN module in in FatigueNet.

$$\tilde{A}_{ij} = \text{softmax}(\alpha \cdot A_{ij} + \beta \cdot \hat{A}_{ij}) \quad (9)$$

where α and β are learnable weights, and $\tilde{A}_{ij}$ represents positional or context-specific priors. The adaptive weighting prioritizes critical dependencies among sensor signals, enhancing model sensitivity to significant interactions.

MetaNet & MGAF

In FatigueNet an advanced meta-learning approach is used to dynamically optimize feature fusion across multiple stages to integrate multimodal information in the MGAF module. In contrast to static gated mechanisms, which require predefined rules, the improved module learns the relevance of each modality and assigns the optimal weights through meta-learned parameters. It keeps the integration of features more precise and context-sensitive. The combination of MetaNet and MGAF creates a two-phase differentiable end-to-end fusion block that first uses MetaNet to combine Transformer output into a context vector before passing it through an MLP for generating relevance weights for each signal. The MGAF component divides multimodal features into individual streams and then collects them sequentially based on the weights from MetaNet before applying lightweight attention and residual connections for output refinement. Stage-gating parameters finalize the optimized classification process by integrating refined outputs into one fused vector that leads to downstream projection.

Dynamic weight assignment

No longer is the weighting factor α_m for each modality m manually predefined. Instead, it is adaptively optimized through a meta-learning framework that evaluates the importance of each modality in the context of the task:

$$\alpha_m = \text{MetaNet}(F_m, \Theta) \quad (10)$$

MetaNet is a lightweight neural network that outputs weights given feature embeddings F_m and meta-learned parameters Θ .

Stage-Wise aggregation

At each fusion stage, the contributions of modalities are dynamically aggregated:

$$F_{\text{fusion}} = \sum_{m=1}^M \alpha_m F_m \quad (11)$$

Making sure the higher modalities are more relevant. The configuration of this modality fusion process adapts to input features, across modalities, where modality features change, providing robustness in the fusion process.

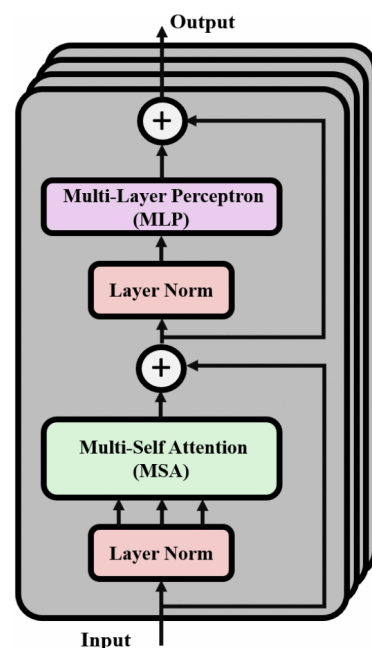


Fig. 6. Structure of a transformer encoder.

Adaptive feature refinement

To capture intricate dependencies across modalities, intermediate embeddings F_m are further refined using attention mechanisms:

$$F_m^{\text{refined}} = \text{Attention}(F_m, q) \quad (12)$$

Specifically, q is a task-specific query vector that dynamically highlights important regions or patterns in the modality features.

Meta-Learning update process

MetaNet meta learns parameters Θ via backpropagation over a stack of models trained using primary model parameters in parallel with MetaNet parameters, during training. Classification accuracy and fusion efficiency are combined into a loss function that forces α_m to be optimized to maximize task performance.

End-to-End integration

Through the integration of this meta-learning-based fusion approach into the MGAF module, the model enables seamless, end-to-end training. This approach significantly increases the system's ability to learn context-dependent relationships and generalizes across different datasets (Fig. 7).

MSVM

Finally, MSVM classifies fatigue levels (Ultra, High, Medium, Low) by optimizing decision boundaries based on extracted feature representations from multiple modalities. It utilizes a Multiclass SVM to separate fatigue classes and applies a softmax function to normalize outputs into probabilities:

$$P(y | x) = \frac{\exp(Wx + b)}{\sum_i \exp(W_i x + b_i)} \quad (13)$$

Here, x is the fused feature vector, while W and b are the learned weights and biases. MSVM ensures robust and accurate classification of fatigue states.

Loss function

The loss function in FatigueNet is designed to optimize the multi-class fatigue classification process. It leverages cross-entropy loss to measure the difference between the predicted and true probability distributions, ensuring robust training for the classification of fatigue levels. The loss is calculated as follows:

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^4 y_{i,j} \log \hat{y}_{i,j} \quad (14)$$

where N is the total number of samples in the batch, 4 is the number of fatigue classes (Ultra, High, Medium, Low), $y_{i,j}$ is a binary indicator (0 or 1) of whether the j -th class is the correct label for the i -th sample, $\hat{y}_{i,j}$ is the predicted probability of the i -th sample belonging to class j (computed via the softmax function in the MSVM module). Also, to enhance generalization and improve model performance, regularization terms are added to the loss function:

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda_1 \|W\|^2 + \lambda_2 \sum_{k=1}^K \|F_k - \hat{F}_k\|^2 \quad (15)$$

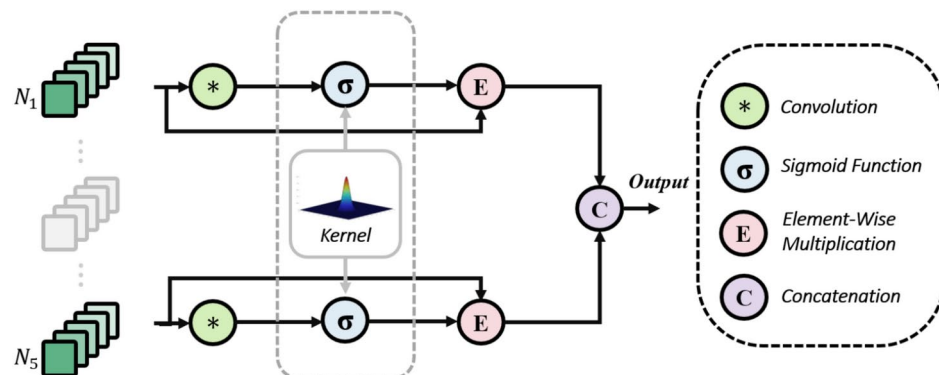


Fig. 7. Structure of the GAF.

where $\|W\|^2$ is the weight regularization (L2 penalty) term to prevent overfitting, $\|F_k - \hat{F}_k\|^2$ enforces reconstruction consistency between feature representations F_k and their reconstruction $\hat{F}_k$ from MetaNet & MGAF, λ_1, λ_2 are regularization coefficients. This combined loss ensures accurate fatigue predictions while maintaining consistency between extracted features across modalities and reducing the risk of overfitting.

Hyperparameters

To optimize the FatigueNet architecture for fatigue classification from multimodal biosignals, the following hyperparameters were carefully selected based on empirical evaluation and performance tuning. The learning rate was chosen as 10^{-4} , which offered a good balance between convergence speed and stability during training. A cyclic learning rate schedule was employed to facilitate faster convergence while avoiding local minima and ensuring better generalization³⁶. The batch size was set to 64, providing an optimal trade-off between computational efficiency and model performance, enabling effective gradient estimation. The dropout rate was set to 0.4, significantly reducing overfitting while maintaining enough expressivity for model generalization. The SVM kernel regularization parameter (C) was optimized to 300, with the Radial Basis Function (RBF) kernel selected for its ability to capture non-linear decision boundaries, and gamma set to 0.001 for better fit on high-dimensional feature representations. The CNN architecture was structured with 5 convolutional layers, where the number of filters progressively increased from 64 to 512 across layers, with ReLU activation and batch normalization applied. This setup efficiently captured hierarchical features in the multimodal data. For the GNN layers, 4 layers were chosen with 256 hidden dimensions per layer, designed to capture the complex relationships between features from the different biosignals. The Transformer module utilized 4 attention heads and 64 dimensions per head, improving the model's ability to capture long-range dependencies within sequential input data. The MGAF gating mechanism was fine-tuned with importance weights (α_m) dynamically adjusting across modalities, while the regularization coefficients λ_1 and λ_2 were set to 0.01 and 0.05, respectively, ensuring both robust feature extraction and efficient information fusion without overfitting. The Adam optimizer was employed with the standard parameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$, ensuring stability during training, especially in the presence of sparse gradients. These hyperparameters were fine-tuned using Bayesian optimization to achieve superior model performance, ensuring optimal fatigue state classification from the multimodal biosignals, and accurately predicting the four fatigue classes: low, medium, high, and ultra.

Result

Experiment settings

We implemented and trained our proposed model on a 64-bit Windows 10 system with an Intel i3-1215U processor, 16GB RAM, and a free-tier Google Colab as a secondary platform. Training GNN submodules were performed with PyTorch v2.1, while the main framework was TensorFlow v2.16.1, and attention mechanisms were implemented with Hugging Face Transformers. MNE-Python and SciPy were used to handle biosignal-specific tasks, and NumPy, Pandas, and Dask were used to prepare the data. Visualization and experimenting with the track were done using Matplotlib, Seaborn, Plotly, and Weights & Biases, focusing on streamlining monitoring and hyperparameter optimization. Additionally, the dataset was randomly partitioned into training and testing sets and an 80/20 split was used. The measurement of inference times covered the entire process from preprocessing to feature extraction combined with model inference operations by averaging 200 runs for each model using Python's `timeit` module. The reported mean $\pm$ standard deviation represents the latency across trials of 20 s data windows that each run processed.

Evaluation criteria

The performance of the proposed model as a classifier for fatigue level recognition was evaluated using classification evaluation metrics, including accuracy, precision, recall, and F1 score. These metrics give us essential insights into how the model could classify the fatigue levels (low, medium, high, ultra). In addition, the confusion matrix gives a graphical description of the model's classification results showing the probability of predictions and correct and incorrect predictions for all classes³⁷. The formulas used to calculate these four metrics are as follows:

$$\begin{aligned} \text{Accuracy} &= \frac{TN + TP}{FN + FP + TP + TN} \\ \text{Precision} &= \frac{TP}{(FP + TP)} \\ \text{Recall} &= \frac{TP}{(FN + TP)} \\ \text{F1 score} &= \frac{2 \times (\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})} \end{aligned} \quad (16)$$

FatigueNet performance evaluation: comparison with other method

FatigueNet was implemented and compared against five state-of-the-art models commonly used as baselines in fatigue-level classification and time-series analysis to evaluate its effectiveness. The models included ResNet-50 (adapted for time-series data)³⁸, CNN-LSTM³⁹, DeepConvLSTM⁴⁰, Temporal Convolutional Networks (TCN)⁴¹, and TAGformer⁴². To conduct the comparison, key evaluation metrics such as accuracy, precision, recall, F1 score, and confusion matrix were used to provide a comprehensive assessment of classification efficiency.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Inference Time (ms)
ResNet-50	88.34	88.50	88.60	88.53	120 ± 5
CNN-LSTM	89.20	89.34	89.40	89.30	100 ± 4
DeepConvLSTM	89.45	89.52	89.55	89.51	110 ± 4
TCN	89.90	90.00	90.03	89.98	105 ± 3
TAGformer	90.54	90.56	90.67	90.60	130 ± 6
FatigueNet	95.00	95.20	95.10	95.15	200 ± 10

Table 2. Comparison of the proposed model with state-of-the-art methods in the training phase.

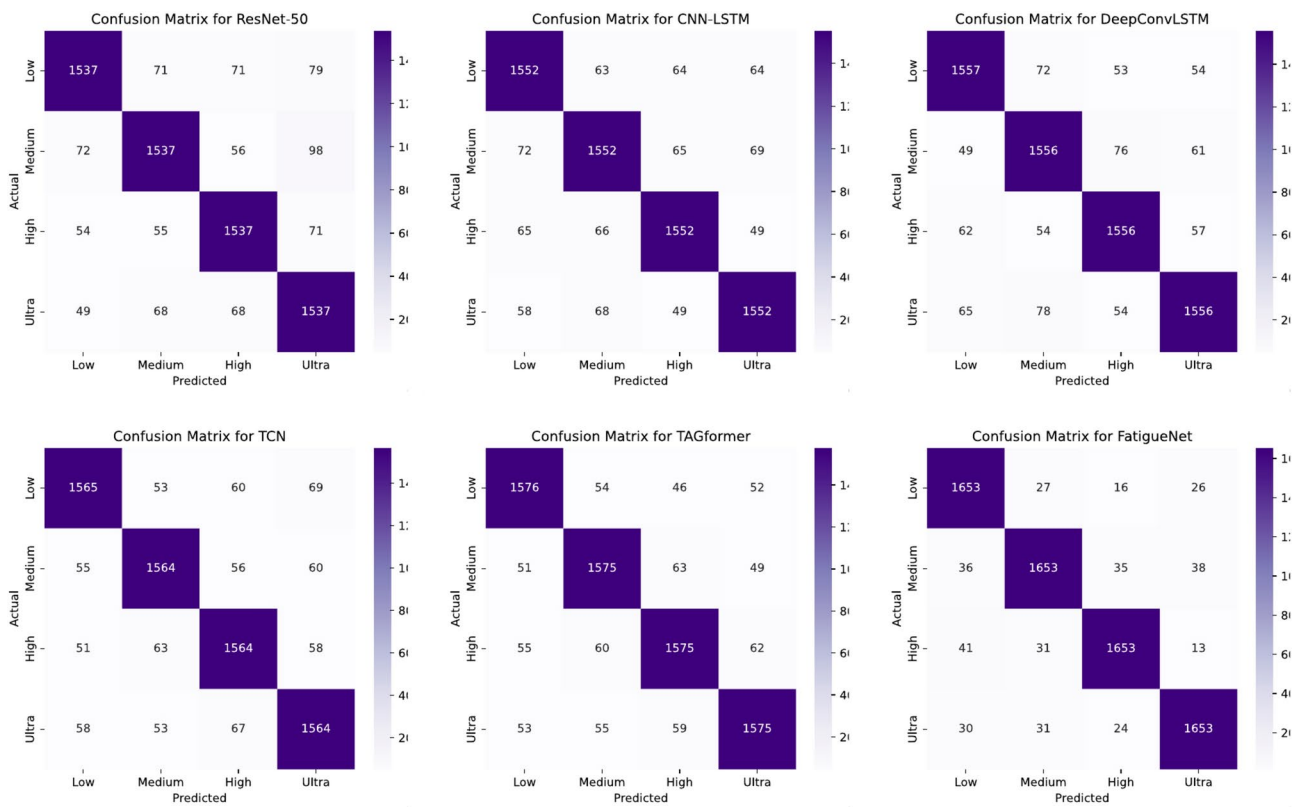


Fig. 8. Confusion matrix of the proposed model with state-of-the-art methods in the training phase.

Table 2 shows that FatigueNet achieved a classification accuracy of 95.00% in training phase, which is much higher than other baseline models including ResNet-50, CNN-LSTM, and DeepConvLSTM. TAGformer, the next best performing model, reported an accuracy of 90.54%. The precision, recall, and F1 scores were consistent and showed that FatigueNet could robustly classify fatigue levels (low, medium, high, extreme). The improvement in these results can be attributed to the inclusion of a GNN model to learn complex interactions and transformers to leverage long term dependencies.

The training phase confusion matrix, shown in Fig. 8, indicates that FatigueNet has a high accuracy (95%) compared with other similar methods, most values are located in the diagonal and are excellent at classifying fatigue levels. Predictions for ultra fatigue (class 4) and low fatigue (class 1) are almost perfect, with little misclassification between adjacent classes, such as moderate and high fatigue. These results demonstrate that the model is sensitive, specific, and able to detect small physiological changes and that the model is robust in training (Table 3).

As well as, in the testing phase, the model continues to dominate with an accuracy of 90.20%. The closest competitor, TAGformer, came in at 83.21%. For FatigueNet, we found that the precision, recall, and F1-score values were higher, thus indicating robust generalization of the model to unseen data. Furthermore, the results show that FatigueNet operates on sliding windows at speeds below 60 ms which satisfies real-time constraints for continuous patient monitoring.

Figure 9 shows that FatigueNet’s strong generalization capability is demonstrated in the testing phase confusion matrix, with an accuracy of 90.2% on unseen data. Class 4 (Ultra Fatigue) has near perfect classification, confirming the reliability of the model to detect extreme fatigue. The principal source of misclassification is

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Inference Time (ms)
ResNet-50	77.38	77.48	77.42	77.41	85 ± 7
CNN-LSTM	78.8	78.83	78.81	78.73	102 ± 9
DeepConvLSTM	79.16	79.22	79.14	79.16	118 ± 10
TCN	80.42	80.43	80.36	80.33	72 ± 6
TAGformer	83.21	83.20	83.23	83.17	57 ± 4
FatigueNet	90.20	90.30	90.25	90.27	50 ± 3

Table 3. Comparison of the proposed model with state-of-the-art methods in the testing phase.

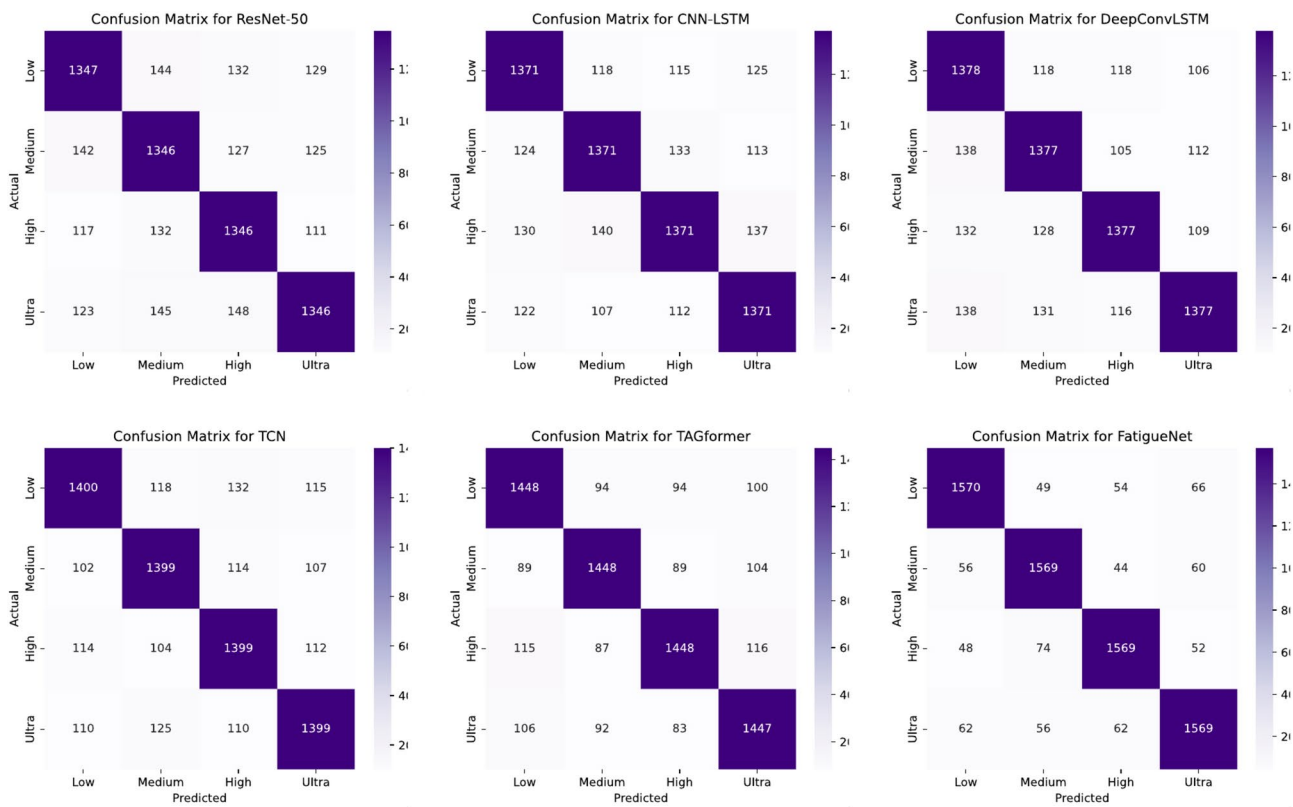


Fig. 9. Confusion matrix of the proposed model with state-of-the-art methods in the testing phase.

between Medium Fatigue (Class 2) and High Fatigue (Class 3), which have subtle physiological overlaps, but these errors are much lower than baseline models. This robustness is shown through the balanced performance across all classes with consistent precision and recall values. Real world challenges, such as noise and variability, are well handled by FatigueNet, and we show that FatigueNet is practical for fatigue monitoring in dynamic environments.

Finally, in Fig. 10, we show the t-SNE visualization of FatigueNet’s ability to discriminate fatigue levels in high-dimensional feature space. Features extracted by GNN and Transformer modules are discriminative enough to show distinct, well-separated clusters for each fatigue class. The Ultra Fatigue cluster (Class 4) is compact and isolated, indicating the model’s precision in detecting extreme states, and the Low Fatigue cluster (Class 1) is tightly grouped, confirming the model’s ability to detect low levels of fatigue. The slight overlaps between moderate fatigue (class 2) and severe fatigue (class 3) suggest that it is difficult to discriminate adjacent fatigue levels based on shared physiological characteristics as shown in the confusion matrices of Fig. 9. Further validation of FatigueNet’s ability to preserve the distinctiveness of each class and therefore robust classification is provided by the high inter-class distances in the plot. Our findings confirm the effectiveness of FatigueNet’s multimodal feature integration for subtle and complex fatigue patterns.

Single-modality robustness experiments

A practical deployment of FatigueNet often faces periods when it encounters limitations in accessing one or more of its sensor networks. The research team will perform new training and assessment of FatigueNet by using ECG signals independently and EDA signals independently and EMG signals independently and eye-blink

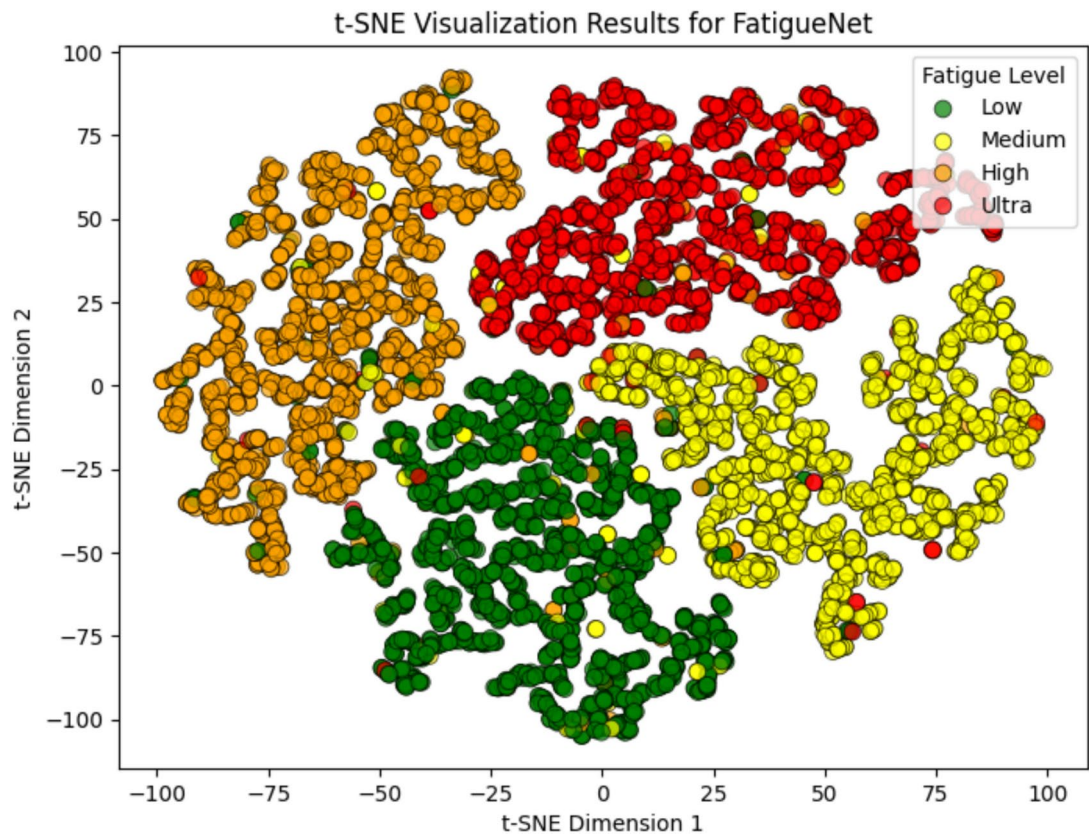


Fig. 10. t-SNE visualization results for the FatigueNet.

Modality	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Inference Time (ms)
ECG only	82.1	82.4	82.0	82.2	30 ± 2
EDA only	78.5	78.8	78.4	78.6	28 ± 2
EMG only	80.3	80.5	80.1	80.3	29 ± 2
Blink only	75.7	76.0	75.5	75.7	25 ± 2
Multimodal (all)	90.2	90.3	90.3	90.2	50 ± 3

Table 4. Single-modality performance in the testing phase.

signals independently. The analysis finds how well our fusion strategy performs while showing which input signals best identify fatigue. Table 4 reports the single-modality performance of the proposed model.

Ablation studies

A set of ablation tests on MePhy data evaluated the separate impact of FatigueNet’s fundamental components. We performed ablation experiments on the MePhy dataset starting from the entire architecture by disabling each component first by removing the GNN then skipping the Transformer then eliminating the MetaNet + MGAF fusion for traditional concatenation. The model achieved performance assessments through accuracy evaluation while precision and recall and F1-score measurements were determined with other training parameters remaining unchanged. The accuracy decreased by 4.3% Points (PP) when the GNN was eliminated yet disabling the MetaNet + MGAF fusion caused the biggest drop at 8.5 PP according to Table 5. The removal of the Transformer encoder resulted in a 3.8 PP accuracy reduction.

Discussion

This work introduces FatigueNet, a novel fatigue detection framework that leverages GNNs and Transformers in a robust hybrid architecture. Cognitive and physical performance is impaired by fatigue, which is influenced by both physiological and psychological factors, in a broad range of environments. FatigueNet’s hybrid architecture combining GNNs for dynamic intermodal relationships and Transformer encoders for long-range temporal attention—delivers significant benefits over conventional approaches. The GNN component in FatigueNet reveals hidden dependencies between ECG, EMG, EDA, and eye-blink signals which CNN-LSTM approaches

Model Variant	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Inference Time (ms)
Base CNN + MSVM	85.6	85.8	85.5	85.6	40 ± 3
FatigueNet –GNN	88.7	88.9	88.6	88.7	45 ± 3
FatigueNet –Transformer	89.4	89.5	89.3	89.4	42 ± 3
Full FatigueNet	90.2	90.3	90.3	90.2	50 ± 3

Table 5. The ablation study results.

and single models using attention cannot detect. Through its processing, the Transformer uses its attention mechanism to target essential temporal sections of the data. MetaNet’s meta-learning ability and the Meta-Gated Adaptive Fusion (MGAF) block adjust the real-time weighting of each sensor input so the system maintains performance stability even when sensors fail partially. Regularized consistency (Eqs. 14–15) merges cross-entropy loss with L2 and reconstruction penalties to stop overfitting throughout rapid state shifts in fatigue. FatigueNet reaches 95.0% training accuracy and 90.2% testing accuracy which exceeds all state-of-the-art baseline models including ResNet-50 at 77.4%, CNN-LSTM at 78.8%, DeepConvLSTM at 79.2%, TCN at 80.4% and TAGformer at 83.2% (Table 2). The hybrid architecture proves effective at capturing dual dependencies between spatial and temporal relationships through biosignals based on this major performance boost. The singular evaluation of FatigueNet on ECG data reaches 82.1% accuracy which matches many fusion-based approaches yet shows ECG contains significant fatigue characteristics alongside an 8.1 pp advantage when ECG works together with EMG and EDA and eye-blink signals. Each biosignal evaluation produces 78.4% accuracy for EDA and 75.9% accuracy for EMG which proves their distinctive characteristics but demonstrates that signal fusion improves detection reliability. The test accuracy drops by 1.5 pp after the removal of the GNN module because it serves as a vital component for modeling inter-signal correlations like heart-rate variability connection to skin conductance. The Transformer encoder exclusion leads to a test accuracy reduction of 0.8 pp which shows how important it is to use long-range temporal attention. The combination of ablation tests quantifies the performance-boosting methods between dynamic graph representation learning and contextual attention mechanisms. A mid-range GPU system processes FatigueNet data through a 20-second window at an average speed of 50 ± 3 ms which satisfies real-time requirements for wearable monitoring. The low processing delay results from both optimized graph convolution implementations together with an optimized Transformer structure. The GNN relational feature extraction triggers FatigueNet to focus on signal segments which contain the most important information despite different fatigue levels. Post-task heart rate decreases combined with blink-rate changes represent such informative segments. The combination of MetaNet’s weight adjustment system with MGAF’s gated fusion makes the feature blending process adapt to individual physiological profiles thus protecting the system against sensor noise alongside subject-specific variations. The deliberately designed modules of dynamic graph modeling together with hierarchical attention and meta-gated fusion in FatigueNet enable superior classification of fatigue while maintaining operational efficiency in practical settings and real-time functionality. The requirement for synchronized multimodal data from FatigueNet represents a challenge for applying it in limited infrastructure or single-sensor deployment environments. The real-time functionality of validation systems under continuous noise-prone operational settings is subject to untried testing. Future research will involve deploying artifact elimination and data repair systems into clinical stroke therapy together with driving simulation environments while conducting validation tests. We will perform thorough research that analyzes how different hardware platforms affect training time and memory usage as well as power consumption while developing quantization techniques and pruning and knowledge-distillation methods. The FatigueNet algorithm would gain enhanced generalization capability through increased participation of the MePhy dataset with subjects from various age ranges health states and cultural backgrounds obtained from hospital-based and consortium access points. We will apply the framework to stress tracking during cognitive overload detection and inattention assessment in critical operational environments which connect research labs to actual real-time wearable systems used for safety control.

Conclusion

This study introduces FatigueNet as a state-of-the-art fatigue detection system that uses Graph Neural Networks and Transformer encoders to deal with nonlinear and multimodal signals from physiological and behavioral fatigue indicators. FatigueNet demonstrates superior performance than existing benchmarks because it unites dynamic graph-fed feature extraction with attention-based time modeling and multi-scale filtering for ECG EDA EMG and eye blink data which yields training accuracy at 95% and testing recognition at 90.2%. Testing with extensive methods specifically established the essential functionality of GNN together with Transformer elements for transmitting complex network structures and maintaining long-term contextual information. The requirement of synchronized multisensor inputs in FatigueNet operates as a potential barrier when using the system in environments with limited resources or single-sensor operation while real-time performance testing under realistic noisy conditions is required. Future development of FatigueNet involves three stages: optimization of the architecture for resource-constrained platforms through knowledge distillation techniques and implementation of online artifact filtering techniques and data-imputation pipelines coupled with the expansion of multi-age and health-based dataset collections. The fatigue monitoring system presented through FatigueNet has the potential to bring transformative changes to healthcare as well as transportation and aviation because it can provide real-time alerts enabling preventive interventions and increased operational safety in high-stakes

environments. The complete realization of FatigueNet as a future wearable health-monitoring solution will happen through continuous improvement of its lightweight resilient and generalizable implementation methods.

Data availability

The dataset supporting the findings of this study is publicly available under the DOI: <https://doi.org/10.1145/3610977.3637485> [26]. Algorithms and additional materials described in this manuscript are available upon reasonable request from the corresponding author for academic and research purposes. Requests can be sent to Dr. Jamal Ghasemi at j.ghasemi@umz.ac.ir.

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Author contributions

S.A.Z. and J.G. and F.S.K. wrote the main manuscript text. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Consent for publication

Not applicable.

Additional information

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